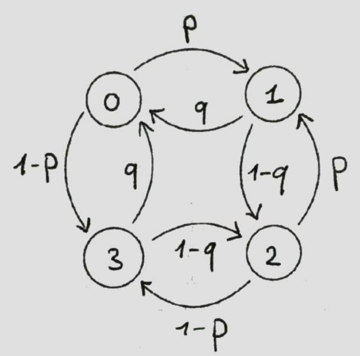
EE 595H : Stochastic Models

A. This is **NOT** an open book exam. The exam is for **25 Marks** and of duration **75 Minutes**. The marks for each question is indicated at the beginning of the question.

B. The question paper consists of **THREE** questions printed on **TWO** A4 sides. Answer to each question should begin on a **FRESH** side of the answer book.

Question 1. (6 marks)

Let $X_0, X_1, X_2, \ldots$ be a homogeneous Markov chain with state space $S = \{0, 1, 2, 3\}$ and transition matrix P represented by the following transition diagram:



1. Compute the transition matrix of the Markov chain.
2. Find all stationary distributions of the Markov chain.
3. Is detailed balance satisfied for the Markov chain?

Question 2. (10 marks)

Let a and b be positive integers, and consider the homogenous Markov chain with the state space

$$\{(x, y) : x \in \{0, \ldots, a-1\},\ y \in \{0, \ldots, b-1\}\},$$

and the following transition mechanism: If the chain is in state (x, y) at time n, then at time $n+1$ it moves to $((x+1) \bmod a, y)$ or $(x, (y+1) \bmod b)$ with probability $\frac{1}{2}$ each.

(a) Show that the Markov chain is irreducible.

(b) Show that the Markov chain is aperiodic if and only if $\gcd(a, b) = 1$.

(c) Find all stationary distributions of the Markov chain.

(d) Is the Markov chain reversible? Can you find an intuitive reason justifying your answer.

Question 3. (9 marks)

We would like to use the Metropolis algorithm to get samples from the binomial distribution $Bin(n, p)$ with parameters n and p. Consider the simple random walk on the state space $S = \{0, 1, 2, \ldots, n\}$ with transition probability $q_{ij} = \frac{1}{2}$ for $|i - j| = 1$, $q_{00} = \frac{1}{2}$ and $q_{nn} = \frac{1}{2}$.

1. Is the given Markov chain appropriate for use in the Metropolis algorithm?
2. What are the acceptance probabilities? Compute them exactly.
3. What are the transition probabilities for the Metropolis chain? Compute them exactly.

END